Water pressure



Materials required:

- Milk carton
- Adhesive tape
- Scissors
- Water jug
- Basin

Procedure:

Punch 3 or 4 holes, one above the other, alongside of an empty milk carton. Cover the wholes with a long strip of adhesive tape and fill the carton with water. Place the carton in a basin and pull of the tape.

From which hole the water will travel farthest ?

Water pressure

Explanation:

The stream from the lowest hole travels farthest and where is why:
The water at the bottom of the carton has the force exerted by the pressure of water above it.
Like air, water has pressure.
Water pressure depends, as your experiment shows, on the water's depth. Many cities pump water into raised tanks. This is done to give the water enough force to run up into people's homes from pipes beneath the ground.

Surface tension



Materials required:

- Needle
- Fork
- Container
- Soap

Procedure:

1. Using a fork as a carrier, place a needle on the surface of the water in a container. Carrefully remove the fork.

Did you manage to make the needle float? Why does it float ?

2. Dip a drop of soap in your container with the floating needle.

What happens with the needle ?

Experiment no. 2

Surface tension

Explanation:

1. The needle is heavier than the ammount of water that it displaces and should be expected to sink. It floats, however, because of an invisible elastic skin. When water comes in contact with air, the molecules on the surface huddle close together and form a thin film or skim over the surface.

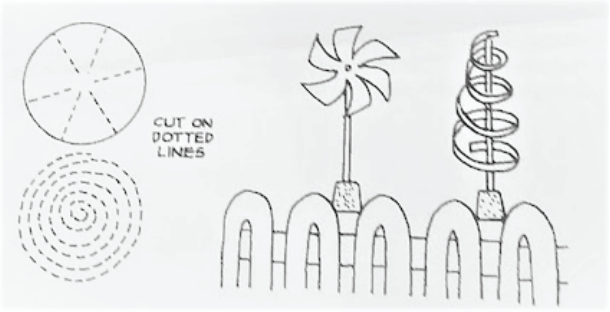
2. In the second part the soap makes the needle sink inmediatly. The soap reduces the surface tension, it is one of the reasons why we use soap for cleaning. By lowering the surface tension, the soap makes the water able to wet greasy surfaces.

Experiment no. 2

How heat travels in air

Materials required:

- Cardboard
- Scissors
- Lamp with a bulb as heat source
- Wooden stick



Procedure:

From a piece of cardboard cur a pin wheel and a spiral as illustrated. Mount each on a wooden stick. Hold each above a hot lighted electric lamp.

What is happening and why ?

How heat travels in air

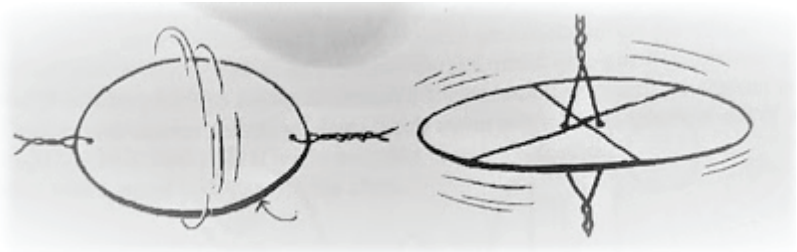
Explanation:

You will see that the pinwheel revolves and the spiral seems to rise. The gadgets are set in motion by air currents produced: the warm air rises and the cold air sinks. Wind is simply: moving air.

Magic colours

Materials required:

- Cardboard
- String
- Scissors
- Paint



Procedure:

1. With water colours or poster paints, colour one side of the cardboard disk red and the other side blue.Punch small holes on the opposite sides of the disk as in the illustration. Thread short lengths of string through each hole. Hold the cardboard by its string and twirl it around.

What colour can you see and why?

2. Make a toy top by dividing a cardboard disk into alternating segments of yellow and blue. Thread a string though the holes in the center as in the illustration. The spin the disk.

What colour can you see and why?

Magic colours

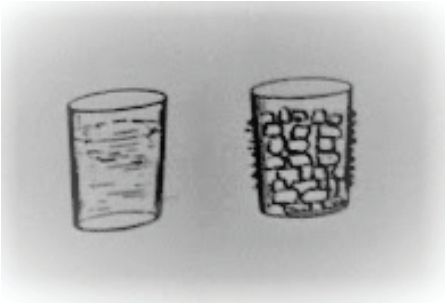
Explanation:

With the first toy you will see purple and with the second you will see green.
The disk reflects both colours. You see a third colour when your eyes and the brain mix the colours of the rapidly whirling disks. This happens because the eye continues to see each colour for a short time after it has disappeared.

Where did the drops of water come from?

Materials required:

- Two glasses
- Ice in a bowl
- Water in a pitcher
- Spoon or tongs



Procedure:

Place both glasses on a table. fill one glass with ice. Fill the other glass with water, Observe the two glasses for 5 minutes.

What did you observe?

Experiment no. 5

Where did the drops of water come from?

Explanation:

TYou will see that drops of water formed on the outside of the glass with ice in it. Moisture in air collects on cool surfaces. This is why there are drops of water on the ourside of glass containing the ice. The water did not come from the glass, it is water from the air around the glass.

Experiment no. 5

What made the mirror foggy?

Materials required:

- Mirror

Procedure:

Hold the mirror with both hands and blow your breath on it.

Did the mirror get foggy? Why?

What made the mirror foggy?

Explanation:

Your breath contains water vapor.
The water vapor from your breath was warm and the mirror cool. When the warm vapor touched the cool mirror the water vapor conddensed in the form of little droplets. That was why the water vapor from your breath made the mirror foggy.

Sinking a cork



Materials required:

- Bowl or beaker
- Drinking glass or small jar
- Cork
- Pitcher with water in it

Procedure:

Fill the large transparent container half full with water. Drop the cork in the water. Lower the drinking glass mouth downward over the cork.

What happens with the cork and why?

Sinking a cork

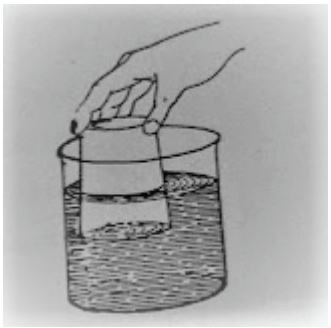
Explanation:

You will see that the cork sinks almost to the bottom of the bowl. When the glass is pushed down in water, the water forces the air into a smaller space.

The air molecules are forced closer together or compressed.

Compressed air pushes the water down. This is why the cork it is pushed almost to the bottom of the bow

What pushed the glass up?



Materials required:

- Bowl or beaker
- Drinking glass
- Pitcher with water in it

Procedure:

Fill the bowl half full of water. Lower the glass mouth downward into the bowl until it touches the bottom of the bowl. Let the glass go.

What happened with the glass and why ?

Experiment no. 8

Water clouds

Explanation:

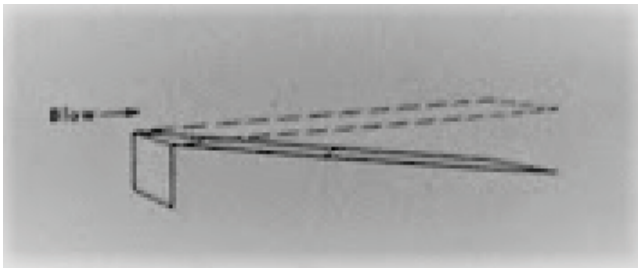
You will see that the glass it is pushed up. When the glass it is pushed down into water, the water forces the air into a smaller space.
The air molecules are forced closer together, or compressed.
When we remove force from the glass, the compressed air pushes the glass back up.

Experiment no. 8

Air streams

Materials required:

- Strip of paper 21 cm long and 4 cm wide



Procedure:

take the strip of paper and fold it 4 cm from one end. Crease the fold well. Hold the short end of the paper strip against your chin with the crease about level with your lips. Blow hard across the top surface of the strip.

Does the paper strip lifts up ?

Air streams

Explanation:

You will see the that the paper strip lifts up. When we blow across the top of the strip, air pressure is reduced at that place. The air pressure below the strip is greater. Air moves from high pressure area toward low pressure area.
That is why the paper strip is pushed up.

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